Major task WG.

Name: AbdelRahman Sherif. Code: 2300100

for TE: $E_y = \begin{cases} A\cos K_x x & -w < x < w \\ Be^{-\gamma x} & x > w \end{cases}$

$$P_{core} = \frac{1}{2Z_{gTE}} * \int_{-w}^{w} A^2 \cos^2 K_x x \, dx \quad \text{"even TE"}$$

$$P_{core} = \frac{1}{2Z_{gTE}} \times A^2 \times \frac{1}{2} \int_{-w}^{w} 1 + \cos(2K_x x)\, dx.$$

$$P_{core} = \frac{2}{2Z_{gTE}} \times \frac{A^2}{2} \int_{0}^{w} (1 + \cos 2K_x x)\, dx$$

$$P_{core} = \frac{A^2}{2Z_{gTE}} \left[w + \frac{\sin(2K_x w)}{2K_x} \right]$$

$$P_{clad} = \frac{2}{2Z_{gTE}} * \int_{+w}^{\infty} B^2 e^{-2\gamma x} dx = \frac{B^2}{Z_{gTE}} \left[\frac{e^{-2\gamma x}}{-2\gamma} \right]_{w}^{\infty}$$

$$P_{clad} = \frac{B^2}{2Z_{gTE}} \left[\frac{e^{-2\gamma w}}{\gamma} \right]$$

$$\Gamma_{TE} = \frac{A^2 \left(w + \frac{1}{2K_x} \sin(2K_x w) \right)}{\frac{B^2 e^{-2\gamma x}}{\gamma} + A^2 \left(w + \frac{1}{2K_x} \sin 2K_x w \right)} \quad (1)$$

from B.Cs $A\cos K_x w = Be^{-\gamma w} \rightarrow \frac{A^2}{B^2} = \frac{e^{-2\gamma w}}{\cos^2 K_x w}$ (2)

- dispersion Relation

$$K_x \tan K_x w = \gamma \quad (3)$$

$$\gamma w = V\sqrt{b} \;, \quad K_x w = V\sqrt{1-b} \quad (4)$$

from (3) & (4) $\tan(K_x w) = \frac{\gamma}{K_x} = \frac{\gamma w}{K_x w} = \frac{V\sqrt{b}}{V\sqrt{1-b}} = \sqrt{\frac{b}{1-b}}$

Name: AbdelRahman Sherif Code: 2300100

$\sin(K_x W) = \sqrt{b}$

$\cos(K_x W) = \sqrt{1-b}$

$$\sin 2K_x W = 2\sin(K_x W)\cos(K_x W)$$
$$= 2\sqrt{b}\sqrt{1-b}$$

$$\Gamma_{TE} = \frac{\left(W + \frac{1}{2K_x} 2\sqrt{b}\sqrt{1-b}\right)}{\frac{\cos^2 K_x W}{\gamma e^{-2\gamma x}} e^{-2\gamma x} + \left(W + \frac{1}{2K_x} 2\sqrt{b}\sqrt{1-b}\right)}$$

$$\Gamma_{TE} = \frac{\gamma W + \frac{\gamma W}{K_x W}\sqrt{b}\sqrt{1-b}}{\cos^2 K_x W + \gamma W + \frac{\gamma W}{K_x W}\sqrt{b}\sqrt{1-b}}$$

$$\Gamma_{TE} = \frac{V\sqrt{b} + \sqrt{\frac{b}{1-b}}\sqrt{b}\sqrt{1-b}}{1-b+V\sqrt{b} + \sqrt{\frac{b}{1-b}}\sqrt{b}\sqrt{1-b}} = \frac{V\sqrt{b}+b}{1-b+V\sqrt{b}+b}$$

$$\Gamma_{TE} = \frac{V+\sqrt{b}}{V+\frac{1}{\sqrt{b}}}$$

TE_{odd} :- $E_y = \begin{cases} A\sin K_x x \\ B e^{\gamma x} \end{cases}$ $\cot(K_x W) = \frac{-\gamma W}{K_x W}$

$$\frac{B}{A} = \frac{\sin K_x W}{e^{-\gamma W}}$$

$$\Gamma_{TE_{odd}} = \frac{\int_{-W}^{W} (A^2 \sin^2 K_x x)\, dx}{2\int_{W}^{\infty} B^2 e^{-2\gamma x} + \int_{-W}^{W} A^2 \sin^2(K_x x)\, dx}$$

Name: AbdelRahman Sherif Code: 2300100

$$\Gamma_{TE_{odd}} = \frac{A^2\left[W - \frac{\sin 2K_x X}{2K_x}\right]}{\frac{B^2 e^{-2\gamma W}}{\gamma} + A^2\left[W - \frac{\sin(2K_x X)}{2K_x}\right]}$$

$$= \frac{A^2(K_x W - \cos K_x W \sin K_x W)}{A^2 \frac{\sin^2 K_x W}{e^{-2\gamma W}} e^{-2\gamma W} * \frac{K_x}{\gamma} + A^2\left[K_x W \cos(K_x W)\sin(K_x W)\right]}$$

$$\Gamma = \frac{V\sqrt{1-b} + \frac{V\sqrt{b}}{V} \cdot \frac{V\sqrt{1-b}}{V}}{\frac{V^2(1-b)}{V^2} \times \frac{\sqrt{1-b}}{\sqrt{b}} + V\sqrt{1-b} + \sqrt{b}\sqrt{1-b}} = \frac{V+\sqrt{b}}{V+1/\sqrt{b}}$$

$$\Gamma_{TE_{even}} = \Gamma_{TE_{odd}}$$

TM even: $H_y = \begin{cases} A\cos K_x X \\ B e^{-\gamma x} \end{cases}$, $E_x = \frac{B}{\epsilon_0 n_i} H_y$

$$\tan K_x W = \frac{\eta\gamma}{K_x}$$

$$\frac{B}{A} = \frac{\cos K_x W}{e^{-\gamma W}}$$

$$\Gamma = \frac{\int_{-W}^{W} \cos^2(K_x X)\,dx}{2\int_{W}^{\infty} \frac{B^2}{A^2} e^{-\gamma x}\,dx + \int_{-W}^{W} \cos^2(K_x X)\,dx}$$

$$= \frac{\left[W + \frac{\sin(2K_x W)}{2K_x}\right]\frac{1}{n_1^2}}{\frac{2}{n_2^2}\frac{\cos^2 K_x W}{e^{-2\gamma W}}\frac{e^{-2\gamma W}}{2\gamma} + \left[W + \frac{\sin 2K_x W}{2K_x}\right]\frac{1}{n_1^2}}$$

$$= \frac{V\sqrt{1-b} + \frac{V\eta\sqrt{b}\,V\sqrt{1-b}}{V^2\left[(\eta^2-1)b+1\right]}}{\eta \cdot \frac{V^2(1-b)}{V^2\left[(\eta^2-1)b+1\right]} \cdot \frac{\sqrt{1-b}}{\sqrt{b}} + V\sqrt{1-b} + \frac{\eta\sqrt{b}\sqrt{1-b}}{(\eta^2-1)b+1}}$$

Name: AbdelRahman Sherif code: 2300100

$$\Gamma_{TM\ even} = \frac{V((\eta^2-1)b+1) + \eta\sqrt{b}}{V((\eta^2-1)b+1) + \frac{\eta}{\sqrt{b}}}$$

TM_{odd}:

$$H_y = \begin{matrix} A\sin K_x w \\ Be^{-\gamma x} \\ -Be^{\gamma x} \end{matrix}, \quad E_x = Z_g^{TM} H_y = \frac{\beta}{\epsilon_0 n_1^2} H_y$$

$$\Gamma_{TH_{odd}} = \frac{\int_{-w}^{w} \frac{A^2}{n_1^2}\sin^2(K_x x)dx}{2\int_{w}^{\infty} \frac{B^2}{n_2^2} e^{-2\gamma x}dx + \int_{-w}^{w} \frac{A^2}{n_1^2}\sin^2(K_x x)dx} = \frac{P_{core}}{P_{clad}+P_{core}}$$

$$\Gamma_{TH} = \frac{\frac{A^2}{n_1^2}\left[w - \frac{\sin(2K_x x)}{2K_x}\right]}{\frac{B^2}{n_2^2}\frac{e^{-\gamma w}}{\gamma} + \frac{A^2}{n_1^2}\left[w - \frac{\sin(2K_x w)}{2K_x}\right]}$$

$$\frac{B}{A} = \frac{\sin(K_x w)}{e^{-\gamma w}}, \quad \cot K_x w = \frac{-\eta\gamma w}{K_x w}$$

$$\Gamma_{TM_{odd}} = \frac{V\sqrt{1-b} + \frac{\eta\sqrt{b}\,V\sqrt{1-b}}{V^2((\eta^2-1)b+1)}}{\eta\frac{V^2(1-b)}{V^2((\eta^2-1)b+1)}\frac{\sqrt{1-b}}{\sqrt{b}} + V\sqrt{1-b} + \frac{\eta\sqrt{b}\sqrt{1-b}}{(\eta^2-1)b+1}}$$

$$\Gamma_{TH_{odd}} = \frac{V[(\eta^2-1)b+1] + \eta\sqrt{b}}{V[(\eta^2-1)b+1] + \eta/\sqrt{b}}$$

$$\Gamma_{TH_{even}} = \Gamma_{TM_{odd}}$$

```
% Slab Waveguide Confinement Factor - Separate Graphs for Each Eta
clear; clc; close all;

n1 = 1.5;
n2 = 1.4;
eta_values = [1.0, 1.070, 1.14]; % 1.0 is TE, >1 is TM
m_values = [0, 1, 2];            % Modes m = 0, 1, 2
V_max = 10;                      % Max V-number for the x-axis

colors = {'b', 'r', 'g'}; % Blue for m=0, Red for m=1, Green for m=2

for e_idx = 1:length(eta_values)
    eta = eta_values(e_idx);

    % Create a brand new figure window for each eta value
    figure;
    hold on;
    legend_entries = {};

    for m_idx = 1:length(m_values)
        m = m_values(m_idx);

        % Sweep U between its strict physical limits for mode m
        % Adding 1e-4 prevents dividing by zero at the exact boundaries
        U_min = m * pi / 2 + 1e-4;
        U_max = (m + 1) * pi / 2 - 1e-4;

        U = linspace(U_min, U_max, 1000);

        % 1. Calculate W directly from the dispersion relation
        W = eta .* U .* tan(U - m * pi / 2);

        % 2. Calculate V directly
        V = sqrt(U.^2 + W.^2);

        % 3. Filter the arrays to only plot up to V_max = 10
        valid_idx = V <= V_max;
        U_valid = U(valid_idx);
```

```
        U_valid = U(valid_idx);
        W_valid = W(valid_idx);
        V_valid = V(valid_idx);

        if isempty(V_valid)
            continue;
        end

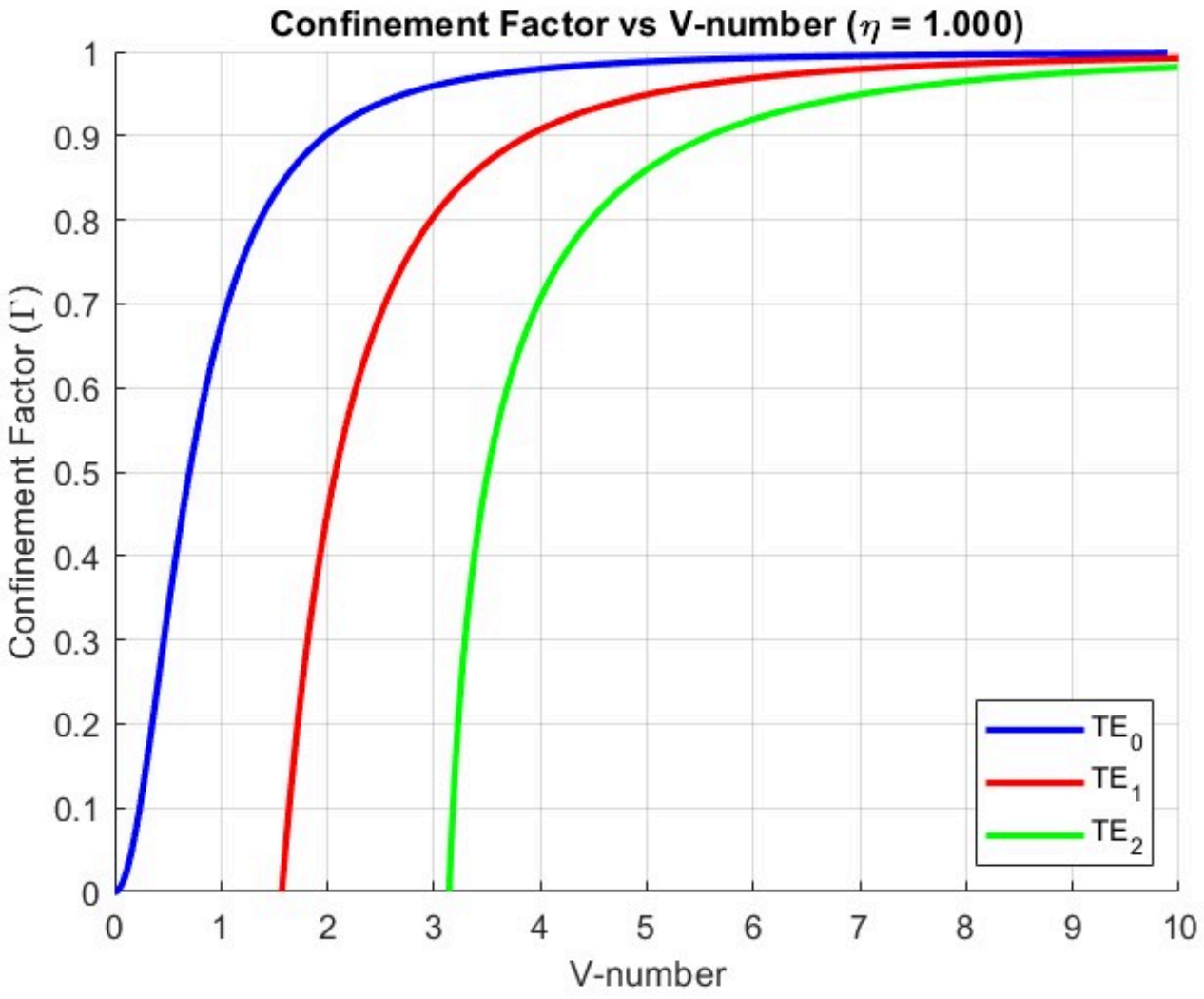
        % 4. Calculate Confinement Factor (Gamma)
        P_core = 1 + cos(m*pi) .* sin(2*U_valid) ./ (2*U_valid);
        P_clad = (cos(U_valid - m*pi/2)).^2 ./ W_valid;

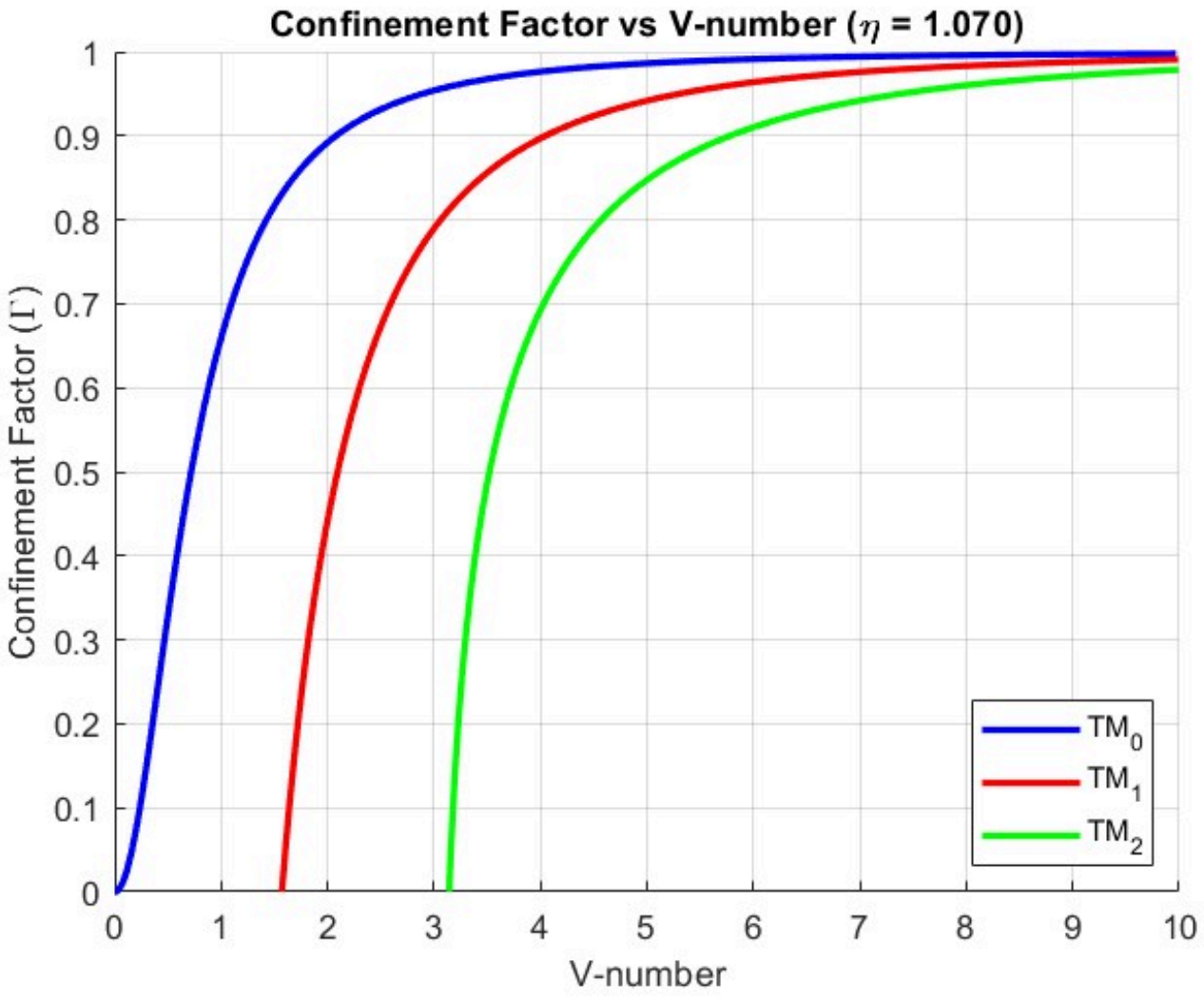
        Gamma = P_core ./ (P_core + eta .* P_clad);

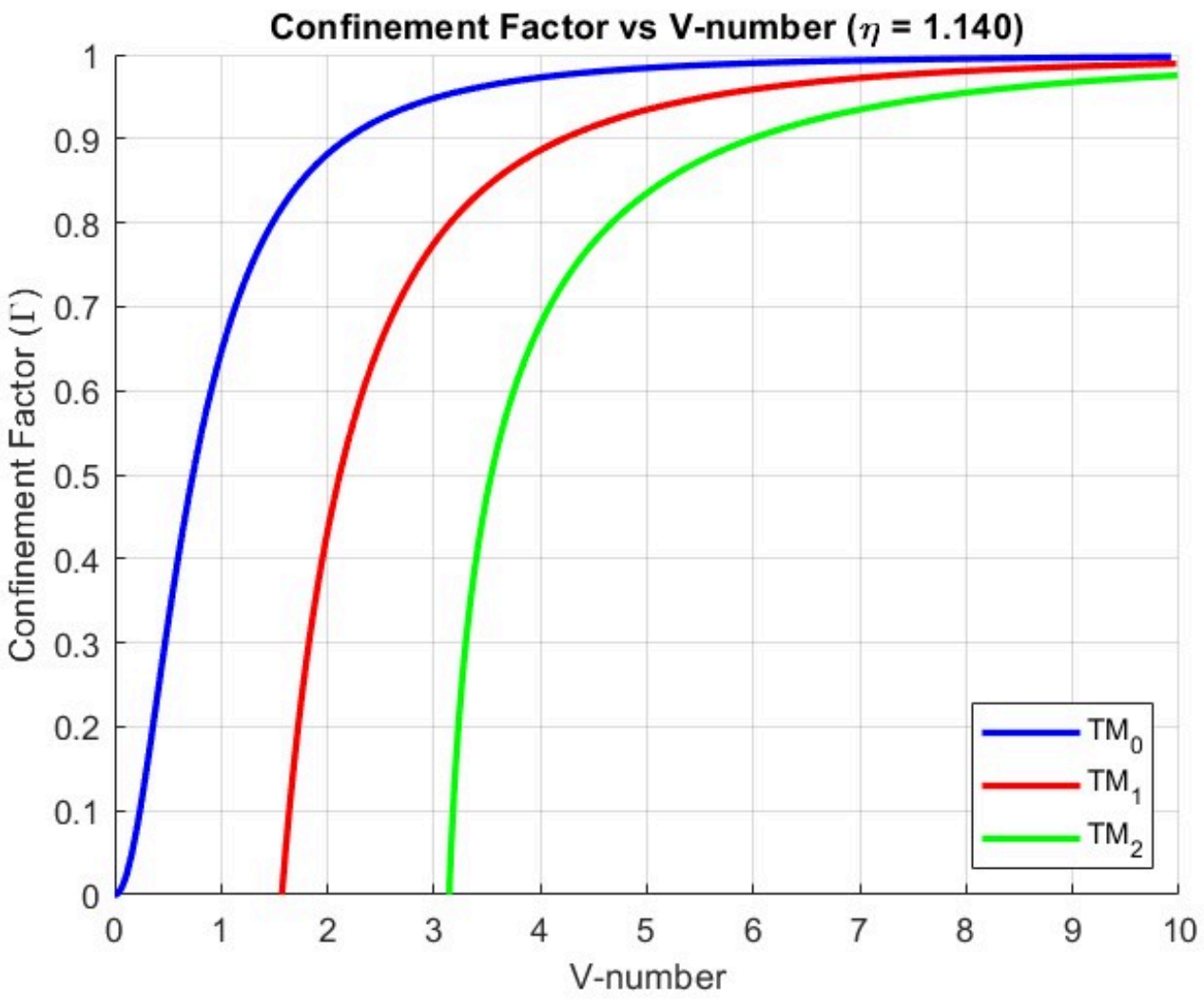
        % Plot the curve
        plot(V_valid, Gamma, 'Color', colors{m_idx}, 'LineWidth', 2);

        % Format legend strings based on TE vs TM
        if eta == 1.0
            mode_str = 'TE';
        else
            mode_str = 'TM';
        end
        legend_entries{end+1} = sprintf('%s_%d', mode_str, m);
    end

    % Graph formatting for this specific figure
    grid on;
    ylim([0 1]);
    xlim([0 V_max]);
    xlabel('V-number');
    ylabel('Confinement Factor (\Gamma)');
    title(sprintf('Confinement Factor vs V-number (\\eta = %.3f)', eta));
    legend(legend_entries, 'Location', 'southeast');
end
```


Confinement Factor vs V-number (η = 1.000)
Confinement Factor (Γ)
V-number
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
0
1
2
3
4
5
6
7
8
9
10
TE$_0$
TE$_1$
TE$_2$


Confinement Factor vs V-number (η = 1.070)
Confinement Factor (Γ)
V-number
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
0
1
2
3
4
5
6
7
8
9
10
TM$_0$
TM$_1$
TM$_2$


Confinement Factor vs V-number (η = 1.140)
Confinement Factor (Γ)
V-number
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
0
1
2
3
4
5
6
7
8
9
10
TM$_0$
TM$_1$
TM$_2$